

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

November 2012

EC 308 Antenna & Wave Propagation

Date: 05.11.2012, Monday

Time: 10:00 p.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Calculate the front to back ratio of an antenna in dB which radiates 2 kW in its most optimum direction and 500 watts in opposite direction 1
- (b) Calculate the approximate gain of paraboloid reflector antenna operating at 6GHz, diameter 20 meters and illumination efficiency 54 %. 2
- (c) The radiation resistance of an antenna is  $63 \Omega$ , loss resistance  $7 \Omega$ . If antenna has power gain 16 dB then find directivity. 1
- (d) Difference between isotropic antenna and omni directional antenna 1
- (e) What is the skip distance of a radio wave? 1
- (f) Explain multi path fading 1
- Q - 2 (a) In far field, for a small Loop Antenna derive field equations. Also derive Radiated Power and Radiation Resistance equation. 10
- (b) Sketch the helical geometry with its associated dimension showing relationship between circumference, spacing, turn length and pitch angle of helix 4

OR

- Q - 2 (a) Explain principal of pattern multiplication Using the pattern multiplication principle, explain the radiation pattern of 4 isotropic elements fed in phase, spaced  $\lambda$ . 7
- (b) Design a broadside Dolph-Tschebyscheff array of 4 elements with spacing  $d$  between the elements and with a major to minor lobe ratio of 26 dB. Find the excitation coefficients and form the Array factor. 7
- Q - 3 (a) Starting from retarded currents derive an expression for E and H components for a short dipole antenna if the spherical system is defined in  $r, \theta$  and  $\Phi$ . 10
- (b) Determine the distance from elementary dipole operating at 1Mhz at which radiation field is 6 times the induction field 4

OR

- Q - 3 (a) Explain different layers of ionosphere 4
- (b) Prove that  $R_r = 73 \Omega$  in  $\lambda/2$  antenna 5

(c) Derive the equation which gives the required shape of the Lens Antenna. 5

### SECTION – II

Q – 4 (a) Which model is the simplest model to analyze the microstrip patch antenna? 1  
(a) transmission line (b) cavity (c) integral equation/method of moments

(b) Microstrip patch antenna is classified as 1  
(a) narrow band (b) broadband (c) medium band

(c) Match the following 5

|                                                |                           |
|------------------------------------------------|---------------------------|
| a) angular range over which a main lobe exists | i) optimum pattern        |
| b) Skelnoff theorem                            | ii) directivity           |
| c) Babinet's Principle                         | iii) beamwidth            |
| d) Tchebycheff Polynomial                      | iv) Null directions       |
| e) measure of the directionality of an antenna | v) Complimentary antennas |

Q - 5 (a) Design a circular patch antenna at frequency 10 GHZ using substrate with dielectric constant of 2.2, height of substrate  $h=0.1588\text{cm}$ . 6

(b) Calculate the dimensions of 3 element Yagi Uda antenna operating at 100 MHz 4

(c) Compare V antenna and Rhombic antenna. 4

### OR

Q - 5 (a) Explain the design steps of log periodic antenna with help of example 5

(b) Explain Babinet's Principle. And discuss it for complementary antenna 4

(c) Explain folded dipole antenna and compare it with half wave dipole 5

Q - 6 (a) Explain the experimental set ups for the measurement of gain of antenna by comparison 5

(b) Describe various types of horn used at microwave frequency 7

(c) Explain in Explain selective fading 2

### OR

Q - 6 (a) A parabolic antenna having circular mouth is to have gain of 1000 at  $\lambda=10\text{cm}$ . Find the diameter of the mouth and half power beam width 5

(b) Differentiate cassegrain feed and offset feed system in reflector antenna 4

(c) Explain the slotted line carriage (SWR) for impedance measurement of antenna under test 5